ORIGINAL ARTICLE

Efficacy and Safety of *KRAS G12C* Inhibitor IBI351 Monotherapy in Patients With Advanced NSCLC: Results From a Phase 2 Pivotal Study

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ABSTRACT

Introduction: *KRAS glycine-to-cysteine substitution at codon 12 (G12C)* mutation is a well-recognized and increasingly promising therapeutic target with huge unmet clinical needs in NSCLC patients. IBI351 is a potent covalent and irreversible inhibitor of *KRAS G12C*. Here, we present the efficacy and safety of IBI351 from an open-label, single-arm, phase 2 pivotal study.

Methods: Eligible patients with NSCLC with *KRAS G12C* who failed standard therapy were enrolled. IBI351

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was orally administered at a dose of 600 mg twice daily. The primary endpoint was confirmed objective response rate assessed by an independent radiological review committee (IRRC) as per Response Evaluation Criteria in Solid Tumors v1.1. Other endpoints were safety, IRRC-confirmed disease control rate, duration of response, progression-free survival (PFS), and overall survival.

Results: As of December 13, 2023, 116 patients were enrolled (Eastern Cooperative Oncology Group Performance Status 1: 91.4%; brain metastasis: 30.2%; prior treatments with both anti-PD-1 or anti-PD-L1 inhibitors and platinum-based chemotherapy: 84.5%). As per the IRRC assessment, the confirmed objective response rate was 49.1% (95% confidence interval [CI]: 39.7–58.6), and the disease control rate was 90.5% (95% CI: 83.7–95.2). The median duration of response was not reached whereas disease progression or death events occurred in 22 patients (38.6%), and the median PFS was 9.7 months (95% CI: 5.6–11.0). overall survival data was immature. Treatment-related adverse events (TRAEs) occurred in 107 patients (92.2%) whereas 48 patients (41.4%) had equal to or higher than grade three TRAEs. Common TRAEs were anemia (44.8%), increased alanine aminotransferase (28.4%), increased aspartate aminotransferase (27.6%), asthenia (26.7%) and presence of protein in urine (25.0%). TRAEs leading to treatment discontinuation occurred in nine patients (7.8%). In biomarker evaluable patients (n = 95), all patients had positive *KRAS G12C* in tissue whereas 72 patients were blood-positive and 23 were blood-negative for *KRAS G12C*. Patients with *KRAS G12C* in both blood and tissue had higher tumor burden at baseline ($p < 0.05$) and worse PFS ($p < 0.05$). Tumor mutation profiling identified *tumor protein p53* (45.3%), *serine/threonine kinase 11* (*STK11*) (30.5%), and *kelch-like ECH-associated protein 1* (21.1%) as the most common genes co-mutated with *KRAS G12C*. Among 13 genes with mutation frequency equal to or higher than 5%, mutations of six genes (*STK11, kelch-like ECH-associated protein 1, phosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit gamma, DNA polymerase epsilon, SMAD family member 4,* and *BMP/retinoic acid-inducible neural-specific protein 3*) were significantly associated with worse PFS ($p < 0.05$). Mutation in *STK11* was also found to have a significant association with higher tumor burden at baseline and lower response rate ($p < 0.05$).

Conclusions: IBI351 monotherapy demonstrated promising and sustained efficacy with manageable safety, supporting its potential as a new treatment option for *KRAS G12C*-mutant NSCLC.

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Keywords: IBI351; GFH925; NSCLC; KRAS G12C; Chinese patients; Pivotal study

Introduction

Lung cancer is one of the most commonly diagnosed cancers and the leading cause of cancer-related death worldwide and in China.[1,2] NSCLC, mainly including lung adenocarcinoma and squamous NSCLC, represents the most common (80–85%) subtype of lung cancer.[3] Mutations in cancer driver genes play important roles underlying pathogenesis and treatment development of NSCLC.[4] As well-recognized cancer drivers, *KRAS* mutations are frequently observed in different tumors.[3] In NSCLC, the prevalence of *KRAS* mutations was up to 30% in Western patients and 9.8% in Asian patients.[3,5] The most common *KRAS* mutation is the glycine-to-cysteine substitution at codon 12 (p. G12C) which prevents the binding of guanosine diphosphates to *KRAS* and results in a highly activated guanosine triphosphate-bound form of KRAS.[3,5]

Despite numerous efforts over the past decades, *KRAS* was previously considered as an undruggable target until the discovery of an allosteric binding site behind the switch II region in *KRAS G12C*-mutant protein.[6] As such, several covalent inhibitors targeting the cysteine residue at the switch II pocket of the KRAS–guanosine diphosphate complex have been developed.[7] Recently, two KRAS G12C inhibitors namely sotorasib[8] and adagrasib[9] received accelerated approval of the Food and Drug Administration for previously treated patients with NSCLC and the *KRAS G12C* mutation. In China, no *KRAS* inhibitor has been approved despite the huge unmet clinical needs of patients with *KRAS*-mutant NSCLC.

IBI351, also known as GFH925, is a potent covalent and irreversible inhibitor targeting *KRAS G12C*. In previously reported first-in-human phase I studies,[10–13] IBI351 was well-tolerated in patients with advanced solid tumors harboring *KRAS G12C* mutation, demonstrating promising efficacy and durable response in those with previously treated advanced NSCLC and colorectal cancer. Herein, we report an open-label, single-arm, phase 2, pivotal study to evaluate the efficacy and safety of IBI351 monotherapy in Chinese patients with advanced NSCLC harboring *KRAS G12C* mutation.

Methods

Patients

Key inclusion criteria were: (1) pathologically confirmed advanced NSCLC, (2) disease progression after standard therapies ($\leq$ 3 lines) including prior anti-PD-1/PD-L1 immunotherapy or platinum-based chemotherapy, (3) presence of the *KRAS G12C* mutation confirmed by the central lab, (4) at least one measurable lesion according to Response Evaluation Criteria in Solid Tumors (RECIST) v1.1, (5) adequate organ function, and

(6) Eastern Cooperative Oncology Group performance status of 0 or 1. Key exclusion criteria included the presence of significant gastrointestinal diseases that may affect swallowing or absorption of the tablets, the presence of active brain metastasis, previous treatment with KRAS G12C inhibitor, and treatment with chemotherapy, targeted therapy, or other antitumor therapies within 28 days or five half-lives (whichever was shorter) before the first dosing of IBI351.

Study Design

In this open-label, single-arm, phase 2, pivotal study, IBI351 monotherapy was orally administered at 600 mg twice daily until disease progression, intolerable toxicity, or other reasons to discontinue treatment, whichever occurred first. The primary endpoint was the confirmed objective response rate (ORR, defined as the proportion of patients with a confirmed complete or partial response) assessed by the independent radiological review committee (IRRC) per RECIST v1.1. Secondary endpoints included ORR assessed by the investigators, disease control rate (DCR), duration of response (DoR), time to response (TTR), progression-free survival (PFS) assessed by IRRC and investigator, and overall survival (OS).

Study Oversight

The study protocol was approved by the institutional review board and ethics committee of all participating sites. This study was registered at clinicaltrials.gov (NCT05005234) and conducted following Good Clinical Practice Guidelines, the Declaration of Helsinki, and relevant local regulatory policy. Written informed consent documents were required before patient enrollment.

Efficacy Assessment

Tumor response was assessed based on computed tomography (CT) or magnetic resonance imaging by IRRC and investigators according to RECIST v1.1. Contrast-enhanced CT or magnetic resonance imaging was used for known or suspected brain metastases. Bone scans, bone X-rays, or both were used for subjects with known or suspected bone metastases. For the baseline assessment, cranial, thoracic, abdominal, and pelvic imaging was performed within the last 28 days before the first dose of IBI351 whereas the bone scan was performed within the last 42 days before the first dose of IBI351. Tumor response assessment was performed every six weeks during the first 48 weeks and then every 12 weeks until disease progression, initiation of new antitumor treatment, informed consent withdrawal, or other reasons leading to study discontinuation (whichever occurred first).

Safety Assessment

Adverse events (AEs) were coded by the Medical Dictionary for Regulation Activities version 26.0 and graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events version 5.0. Safety assessment was performed from the signing of informed consent till up to 30 days after the last dose of IBI351. Treatment-related AEs (TRAEs) were assessed by investigators. All AEs were followed until resolved, recovered to baseline, stable, or deemed irreversible by the investigator, with reasonable explanation (such as loss to follow-up and death), or confirmed as unrelated to the study drug or study procedures at the end of the safety follow-up period.

Biomarker Analysis

Plasma samples were collected at baseline. The OncoCompass Target circulating tumor DNA panel[14] (Burning Rock Biotech, Guangzhou, China) covering 168 genes was used for acquiring tumor mutation profiling information on somatic alterations including single nucleotide mutation, insertion or deletion mutation, rearrangement (fusion), copy number variation, and microsatellite instability (See details in Supplementary Methods for biomarker analysis). Synonymous mutations were removed for further analysis. The hazard ratio (HR) was calculated by the Cox proportional hazards model in the PFS analysis. Statistical significance was calculated by the chi-squared test in the ORR analysis.

Statistical Analysis

The efficacy of IBI351 would be considered statistically significant if the lower limit of the Clopper-Pearson 95% confidence interval (CI) for ORR was above a benchmark of 23%, which was based on confirmed ORRs reported in previous publications.[15–24] In the present study, the ORR of IBI351 was expected to be 35%. With a type I error $\alpha = 0.025$ (one-tailed), 105 efficacy-evaluable subjects could provide a statistical power of equal to or higher than 80%. Patients who received at least one dose of IBI351 were included for the safety and efficacy analysis. The 95% CI for ORR and DCR was calculated using the Clopper-Pearson method. Time-to-event endpoints, such as DoR, PFS, and OS, were summarized with Kaplan-Meier estimates and 95% CIs. Prespecified subgroup analysis of IRRC-assessed ORR was performed in patients with different baseline characteristics including sex, age, smoking, Eastern Cooperative Oncology Group performance status, brain metastasis, liver metastasis, bone metastasis, adrenal metastasis, prior lines of treatment, prior treatments with anti-PD-1/PD-L1 inhibitor, and/or platinum-based

chemotherapies. Statistical analysis was performed using SAS version 9.4 and R version 4.3.2.[25]

Results

Patients

As of December 13, 2023, 193 patients were screened and 116 patients were enrolled and received IBI351 monotherapy (Supplementary Fig. 1). The median follow-up was 15.1 months (95% CI: 14.8–15.7). As of the cutoff date, the median treatment duration of IBI351 was 201 days (range: 11–525), and 31 (26.7%) patients were still on treatment. Patient characteristics at baseline are summarized in Table 1. The median age was 63 years (range: 45–84), and 87.9% were males, and 79.3% were past or current smokers. Most patients had adenocarcinoma (96.6%), and most had stage IV disease (96.6%) with common metastasis sites including bone (43.1%), brain (30.2%), and liver (18.1%). Of the patients, 41.4% had equal to or higher than two prior lines of systemic anticancer therapy, and 84.5% of patients had received both anti-PD-1 or PD-L1 immunotherapy and platinum-based chemotherapy.

Table 1. Patient Characteristics at Baseline

Patients	N = 116
Gender, n (%)	
Male	102 (87.9)
Female	14 (12.1)
Age, y	
median (range)	63 (45-84)
ECOG PS, n (%)	
0	10 (8.6)
1	106 (91.4)
Smoking, n (%)	
Never	24 (20.7)
Past	81 (69.8)
Current	11 (9.5)
Pathological subtype, n (%)	
Adenocarcinoma	112 (96.6)
Squamous cell carcinoma	1 (0.9)
Others	3 (2.6)
TNM stage, n (%)	
III	4 (3.4)
IV	112 (96.6)
Site of metastasis, n (%)	
Bone	50 (43.1)
Brain	35 (30.2)
Liver	21(18.1)
Prior lines of treatment, n (%)	
1	68 (58.6)
≥2	48 (41.4)
Prior treatments, n (%)	
Anti-PD-1 or PD-L1	101 (87.1)
Platinum-based chemotherapy	113 (97.4)
Both	98 (84.5)

ECOG PS, Eastern Cooperative Oncology Group Performance Status; N, number of patients; PD-1, programmed death-1; PD-L1, programmed death-ligand 1; TNM, tumor node metastasis.

Efficacy

Among all patients included in the efficacy analysis (n = 116), 71 patients had the best overall response of partial response with an unconfirmed ORR of 61.2% (95% CI: 51.7–70.1, Table 2), and 57 patients had a confirmed response with confirmed ORR of 49.1% (95% CI: 39.7–58.6) as assessed by IRRC (Fig. 1). The 95% CI of the confirmed ORR by IRRC had a lower limit of 39.7%, which exceeded the prespecified benchmark (23%) for statistical significance.

For secondary endpoints, the confirmed ORR assessed by the investigator was 48.3% (95% CI: 38.9–57.7). As summarized in Table 2, the IRRC-assessed DCR was 90.5% (95% CI: 83.7–95.2). In 57 patients with a confirmed objective response as assessed by IRRC, the median TTR was 1.4 months (range: 1.2–9.7), and the median DoR was not reached whereas 22 patients (38.6%) had events; 18 patients had disease progression and four patients died. The 12-month DoR rate was 53.7% (95% CI: 38.0–67.1). The median PFS was 9.7 months (95% CI: 5.6–11.0). The 12-month PFS rate was 37.9% (95% CI: 27.9–47.9) (Fig. 1, Table 2). OS data was immature with a 12-month OS rate of 54.4% (95% CI: 44.9–63.0). The DCR, DoR, TTR, and PFS assessed by the investigator are summarized in Supplementary Table 1. The intracranial activity of IBI351 was assessed by the investigator with intracranial ORR of 22.6% (7/31, all achieved CR) and intracranial DCR of 96.8% (30/31, one patient had no post-baseline tumor assessment).

Subgroup analysis of IRRC-assessed ORR (Fig. 2) indicated that most subgroups could benefit to a similar extent from IBI351 monotherapy. In patients with brain metastasis, the confirmed ORR was 48.6% (95% CI: 31.4–66.0). In 35 patients (30.2%) with brain metastasis at baseline, there were 11 patients with previous radiotherapy and 24 patients without previous radiotherapy. As presented in Supplementary Table 2, comparable confirmed ORR was observed in patients with and without baseline brain metastasis. Numerical longer median DoR, PFS, and OS were observed in patients without baseline brain metastasis. In patients with both prior anti-PD-1/PD-L1 immunotherapy and platinum-based chemotherapy, the confirmed ORR was 49.0% (95% CI: 38.7–59.3).

Safety

In all patients included in the safety analysis (n = 116), TRAEs occurred in 107 patients (92.2%) including 48 patients (41.4%) with at least one equal to or higher than grade three TRAEs. The most common

Table 2. Summary of Efficacy Assessments by IRRC and OS

Patients	600 mg twice daily, N = 116
Best overall response , n(%)	
Partial response	57 (49.1)
Stable disease	48 (41.4)
Progressive disease	7 (6.0)
Not evaluable[a]	2 (1.7)
Missing[b]	2 (1.7)
Confirmed ORR (95% CI)	49.1% (39.7-58.6)
Confirmed DCR (95% CI)	90.5% (83.7-95.2)
DoR	
Median DoR (95% CI), mo	NC (8.3-NC)
12-mo DoR rate (95% CI)	53.7% (38.0-67.1)
TTR	
Median (range), mo	1.38 (1.2-9.7)
PFS	
Median PFS (95% CI), mo	9.7 (5.6-11.0)
12-mo PFS rate (95% CI)	37.9% (27.9-47.9)
OS	
12-mo OS rate (95% CI)	54.4% (44.9-63.0)

[a]Not evaluable: Two patients received tumor assessment less than 35 days after the first dose of IBI351.
[b]Missing: Two patients discontinued the study before their first scheduled tumor response assessment as per the patient's request (n = 1) and intolerable toxicity (n = 1).
CI, confidence interval; DCR, disease control rate; DoR, duration of response; IRRC, independent radiological review committee; NC, not calculable; ORR, objective response rate; OS, overall survival; PFS, progression-free survival; TTR, time to response.

TRAE was anemia (44.8%), followed by increased alanine aminotransferase (28.4%), increased aspartate aminotransferase (27.6%), asthenia (26.7%), and presence of protein in urine (25.0%). Other common TRAEs are listed in Table 3. Anemia in most patients was mild (grade 1) to moderate (grade 2) whereas it was equal to or higher than grade three in 13 patients (11.2%). Other common TRAEs equal to or higher than grade three (incidence $\geq$ 5%) included increased gamma-glutamyl transferase (7.8%), abnormal hepatic function (7.8%), and decreased lymphocyte count (5.2%). Treatment-related gastrointestinal toxicities commonly included vomiting (8.6%), nausea (7.8%), abdominal distension (7.8%), constipation (5.2%), and diarrhea (2.6%). No grade three or higher gastrointestinal toxicities occurred except that one patient had grade three abdominal pain. TRAEs leading to treatment interruption occurred in 44 patients (37.9%), mainly due to anemia (7.8%) and abnormal hepatic function (7.8%). TRAEs leading to dose reduction occurred in 24 patients (20.7%), mainly due to anemia (6.9%) and abnormal hepatic function (3.4%). TRAEs leading to treatment discontinuation occurred in nine patients (7.8%) due to anemia in two patients, abnormal hepatic function in two patients, increased bilirubin conjugate in one patient, interstitial lung disease in one patient, cardiac failure in one patient, and death of two patients. TRAEs leading to death occurred in three patients (2.6%) (Supplementary Table 3).

Biomarker Analysis

In biomarker-evaluable patients (n = 95), baseline characteristics and IBI351 efficacy were comparable with the total patients (Supplementary Tables 4 and 5). All patients for biomarker evaluation had positive *KRAS G12C* in tissue and 72 patients were blood-positive and 23 patients were blood-negative for *KRAS G12C* with a concordance rate of 75.8% (72/95). Patients with blood-positive *KRAS G12C* (n = 72) had worse PFS compared to blood-negative patients (median mo: 5.55 versus NA, HR = 2.99, p = 0.0078, Supplementary Fig. 2*A*). A numerically lower response rate was also observed in blood-positive patients than in blood-negative patients (49.3% versus 72.7%, p = 0.0931, Supplementary Fig. 2*B*). Notably, our study observed that patients with plasma-positive *KRAS G12C* had significantly higher tumor burden at baseline than did plasma-negative *KRAS G12C* (p = 0.0011, Supplementary Fig. 2*C*). Indeed, patients with higher tumor burden at baseline (defined as above the median of 55.5 mm) had significantly worse PFS (HR = 1.81, p = 0.024, Supplementary Fig. 2*D*) and numerically lower response rate (45.5% vs 56.1%, p = 0.3463, Supplementary Fig. 2*E*).

Tumor mutation profiling also identified a total of 94 genes co-mutated with *KRAS G12C* (Supplementary Fig. 3*A*). The most common co-mutated genes were *tumor protein p53* (45.3%), *serine/threonine kinase 11* (*STK11*) (30.5%), and *kelch-like ECH-associated protein 1* (*KEAP1*) (21.1%). Among 13 genes with mutation frequency equal to or higher than 5%, mutations of six genes (*STK11, KEAP1, phosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit gamma, DNA polymerase epsilon, SMAD family member 4, and BMP/retinoic acid-inducible neural-specific protein 3*) were significantly associated with worse PFS (adjusted $p < 0.05$, Supplementary Fig. 3*B*, Supplementary Table 6). For these PFS-associated genes, *KEAP1* was highly co-mutated with *STK11* (14/29) in comparison to the other four genes. In addition to PFS, the ORR by status of co-mutated genes was also analyzed (Supplementary Table 6). Patients with *STK11* mutations also had nominally significant lower response rates (35.7% in mutated versus 63.5% in wildtype, $p < 0.05$, adjusted $p > 0.05$, Supplementary Fig. 3*C*). No significant differences were observed for other genes, despite some mutations showing trends toward lower ORR (Supplementary Table 6). All patients with *STK11* mutations were also positive for *KRAS G12C* in their blood. Significant different mutation rates of *STK11* ($p < 0.0001$) were observed in patients with higher

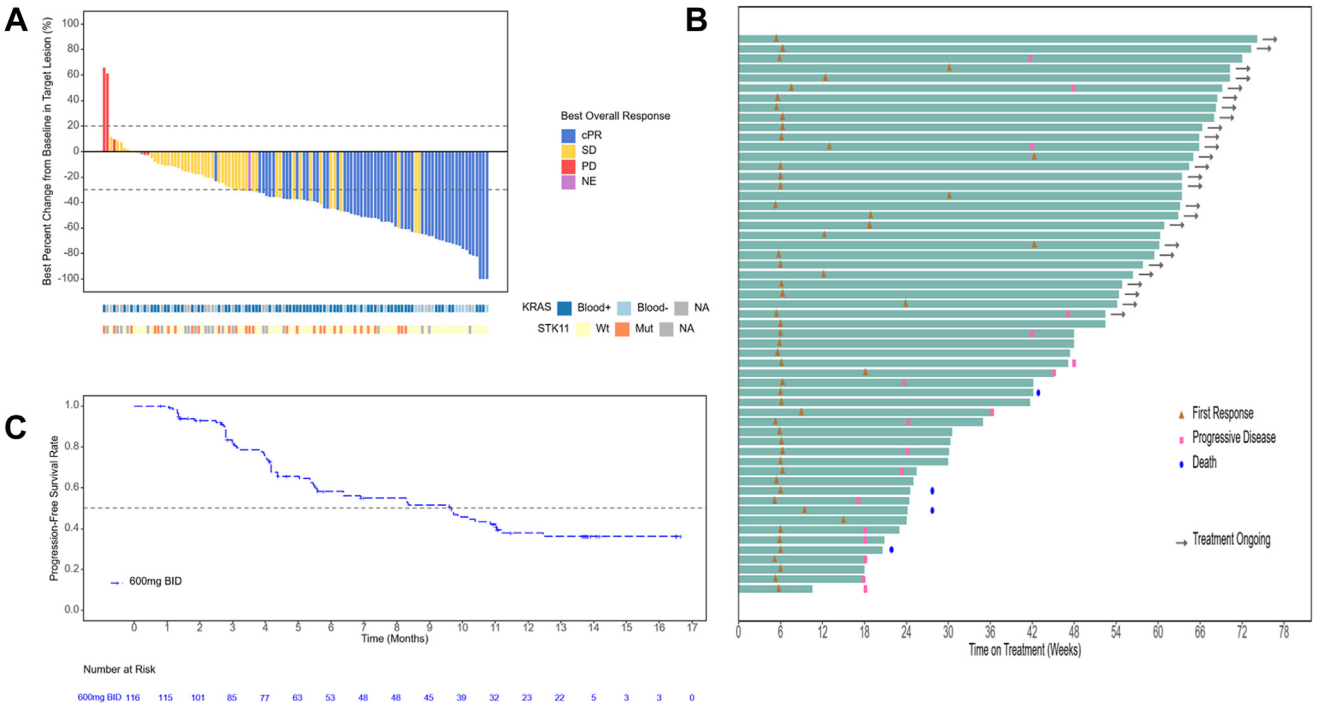

Figure 1. Efficacy of IBI351 in NSCLC patients at 600 mg twice daily by IRRC per RECIST v1.1. *(A)* Best percent change from baseline in target lesions, *(B)* tumor response assessments of each patient during the study, and *(C)* progression-free survival. In Figure 1*A*, one subject with the best response of cPR had the best change in target lesion from the baseline of less than 30%. The IRRC identified one baseline target lesion as an inflammatory non-tumor lesion based on biopsy results, leading to a modification of the overall tumor assessment to PR due to a 42% reduction in another target lesion. Although earlier target lesion assessments could not be changed, the IRRC charter permits adjustments to overall tumor assessment results with a valid rationale. Thus, the plot displays the original target lesion change (a reduction of 23.1%). cPR, confirmed partial response; IRRC, independent radiology review committee; ORR, objective response rate; PD, progressive disease; RECIST, Response Evaluation Criteria in Solid Tumors; SD, stable disease.

tumor burden at baseline (21/46) compared to those with low tumor burden (8/49).

Discussion

In the open-label, single-arm, phase 2, pivotal study, IBI351 monotherapy significantly improved objective response rate with an IRRC-confirmed ORR of 49.1% (95% CI: 39.7–58.6) in 116 patients with previously treated *KRAS G12C*-mutant NSCLC. This is the first pivotal study of a KRAS G12C inhibitor, to the best of our knowledge, to show superior ORR in a large sample size of Chinese patients with NSCLC, with efficacy benefits comparable to global studies of approved KRAS G12C inhibitors, including sotorasib and adagrasib. In the pivotal phase 2 trials of sotorasib and adagrasib in these targetted patients, the confirmed ORR was 37.1% with sotorasib[8] (960 mg, once daily) based on 124 efficacy-evaluable patients and was 42.9% with adagrasib[9] (600 mg, twice daily) based on 112 evaluable patients. Caution should be exercised when comparing across studies considering different patients' baseline characteristics, previous treatment, study design, etc. Nevertheless, in addition to ORR, this study also reported promising DoR and PFS benefits. For IBI351, the median DoR was immature with a 12-month DoR rate of 53.7% (95% CI: 38.0–67.1). The median DoR was 11.1 months for sotorasib[8] and 8.5 months for adagrasib.[9] The median PFS of IBI351 was numerically more favorable than sotorasib[8] and adagrasib[9] (9.7 [95% CI: 5.6–11.0] versus 6.8 versus 6.5 mo, respectively). In our study, OS data was immature with a 12-month OS rate of 54.4% (95% CI: 44.9–63.0). For other *KRAS G12C* inhibitors, the median OS was 12.5 months for sotorasib[8] and 12.6 months for adagrasib.[9]

In advanced NSCLC, brain metastases occurred frequently and resulted in poor prognoses.[26] Our study enrolled a higher proportion of patients with baseline brain metastasis (30.2%) than studies of sotorasib[8] (20.6% brain metastasis at baseline) and adagrasib[9] (20.7% central nervous system [CNS] metastasis at baseline). During post-hoc analysis, the ORR of IBI351 was comparable between patients with and without baseline brain metastasis (48.6% and 49.4%, respectively). Patients without baseline brain metastasis reported numerically longer PFS (median: 10.4 and 6.4

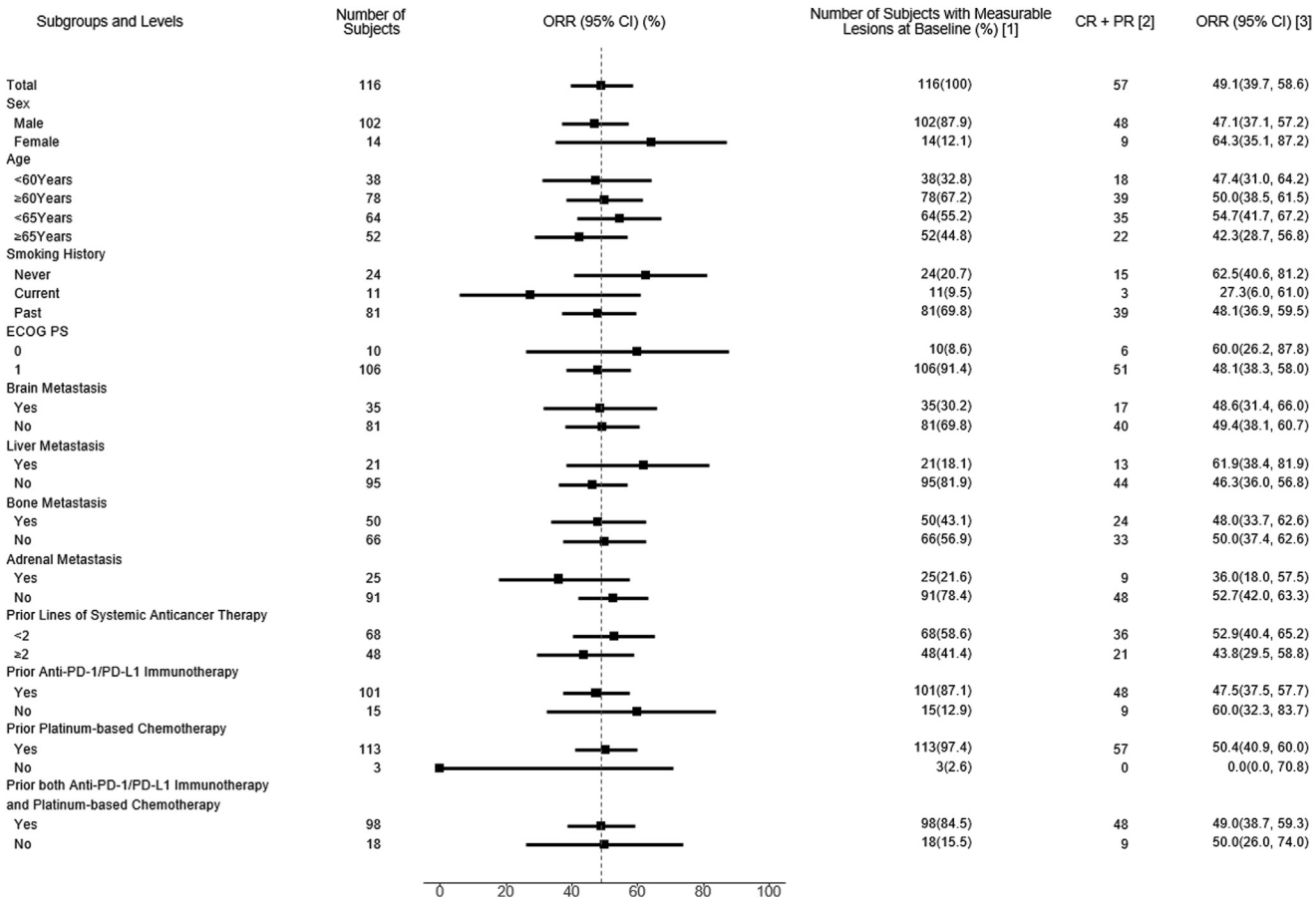

Figure 2. Subgroup objective response rates in NSCLC patients at 600 mg twice daily by IRRC per RECIST v1.1. [1] The denominator for percentage calculation is the number of subjects with measurable lesions at baseline assessed by IRRC according to RECIST v1.1. [2] Evaluation conducted in subjects with measurable lesions at baseline assessed by IRRC according to RECIST v1.1. [3] Calculation of 95% CI using the Clopper-Pearson method in subjects with measurable lesions at baseline assessed by IRRC according to RECIST v1.1. CI, confidence interval; CR, complete response; ECOG PS, Eastern Cooperative Oncology Group Performance Status; IRRC, independent radiology review committee; ORR, objective response rate; PD-1, programmed death-1; PD-L1, programmed death-ligand 1; PR, partial response; RECIST, Response Evaluation Criteria in Solid Tumors.

mo, respectively) and OS (median: 14.1 and 10.9 mo, respectively) than those with baseline brain metastasis. In the phase 2 study of adagrasib in previously treated NSCLC, subgroup ORR was 50.0% in patients with baseline CNS metastasis and 39.7% in patients without CNS metastasis.[9] The intracranial ORR of IBI351 in our study was 22.6% (all achieved CR) which was comparable to other KRAS G12C inhibitors including sotorasib (13%)[27] and adagrasib (33%)[9] in NSCLC patients with previously treated and stable CNS metastases.

Safety profiles of IBI351 were manageable and comparable with the previously reported phase I studies[10–13] and no new safety signal was observed. The most common TRAE of IBI351 was anemia which occurred in 44.8% of patients with events of grade three or higher in 11.2% of patients. Most grade three or higher cases of anemia could be managed by temporary treatment interruption or dose reduction and only two patients discontinued IBI351 treatment. We observed a higher incidence of grade three or higher anemia in patients with a history of anemia at baseline compared with those without a similar history (15.8% versus 2.4%). In addition, the preclinical study of IBI351 observed reduced hemoglobin levels in vivo (data not shown), indicating a need for further exploration into the mechanisms underlying anemia. Given that most clinical trials of KRAS G12C inhibitors were performed in Western populations and our study focused on Chinese patients, it is also important to uncover potential ethnic differences between these groups. A similar incidence of anemia was also observed in adagrasib[9] (36.2% for any grade and 14.7% for grade ≥ 3). Diarrhea and nausea were reported as the most frequent TRAEs in sotorasib[8] (31.7% and 19.0%) and in adagrasib[9] (62.9% and 62.1%). Whereas, IBI351 had a relatively lower incidence of treatment-related gastrointestinal toxicities including vomiting

Table 3. Summary of Treatment-Related Adverse Events

Adverse Events, n (%)	Any Grade	Grade ≥ 3
TRAE	107 (92.2)	48 (41.4)
TRAE leading to treatment interruption	44 (37.9)	39 (33.6)
TRAE leading to dose reduction	24 (20.7)	8 (6.9)
TRAE leading to treatment discontinuation	9 (7.8)	7 (6.0)
TRAEs leading to death	3 (2.6%)	3 (2.6%)
Common TRAEs (≥10% or G3 ≥ 5%)		
Anemia	52 (44.8)	13 (11.2)
Alanine aminotransferase increased	33 (28.4)	3 (2.6)
Aspartate aminotransferase increased	32 (27.6)	1 (0.9)
Asthenia	31 (26.7)	4 (3.4)
Protein urine present	29 (25.0)	0
Blood bilirubin increased	26 (22.4)	3 (2.6)
Edema peripheral	22 (19.0)	0
Pruritus	21 (18.1)	2 (1.7)
Hypoalbuminemia	20 (17.2)	0
Gamma-glutamyl transferase increased	19 (16.4)	9 (7.8)
Hypoesthesia	17 (14.7)	0
White blood cell count decreased	17 (14.7)	2 (1.7)
Rash	16 (13.8)	2 (1.7)
Blood lactate dehydrogenase increased	16 (13.8)	0
Blood alkaline phosphatase increased	16 (13.8)	3 (2.6)
Amylase increased	14 (12.1)	2 (1.7)
Lymphocyte count decreased	13 (11.2)	6 (5.2)
Neutrophil count decreased	12 (10.3)	3 (2.6)
Hepatic function abnormal	10 (8.6)	9 (7.8)

G3, grade 3; TRAE, treatment-related adverse events.

(8.6%), nausea (7.8%), abdominal distension (7.8%), constipation (5.2%), and diarrhea (2.6%), all being grade 1to 2.

Release of circulating tumor DNA into peripheral blood has been associated with tumor burden and is indicative of advanced or metastatic disease.[28] Comprehensive profiling of relevant biomarkers in both tissue and blood could provide additional insights for evaluating personalized treatment responses. Indeed, our study observed that the presence of *KRAS G12C* in both blood and tissue was associated with higher tumor burden at baseline ($p < 0.0001$) and worse PFS (HR = 2.99, $p = 0.0078$). We also identified several genes co-mutated with *KRAS G12C* and observed frequent mutations in *tumor protein p53*, *STK11,* and *KEAP1*, which were consistent with previous studies of other *KRAS G12C* inhibitors.[8,9] *KEAP1* was highly co-mutated with *STK11*. Significant associations between PFS and six genes (including *STK11, KEAP1, phosphatidylinositol-4,5-bisphosphate 3-kinase catalytic subunit gamma, DNA polymerase epsilon, SMAD family member 4,* and *BMP/retinoic acid-inducible neural-specific protein 3*) were observed ($p < 0.05$). Mutation of *STK11* was significantly associated with lower PR rates (35.7% in mutated versus 63.5% in wild type, $p = 0.0258$). Unlike *KRAS, STK11* is a tumor suppressor gene that acts as a negative regulator of mammalian targets for rapamycin signaling.[29] The prognostic role of *STK11* mutations in NSCLC remains undefined with mixed results although it was reported that co-mutations of *KRAS* and *STK11* may predict resistance to immune checkpoint inhibitors.[30] In our study, we observed significant associations of *STK11* mutations with *KRAS G12C* in blood and higher tumor burden at baseline. The role of *STK11* mutations underlying treatment with *KRAS G12C* inhibitors remains to be further characterized.

Potential limitations of our study should also be noted. As a single-arm study, there lacks a control group to compare the efficacy and safety of IBI351 with current treatment options for NSCLC. In a recent randomized, open-label, phase 3 trial, sotorasib was found to significantly improved PFS and favorable safety profiles, compared with docetaxel, in previously treated patients with NSCLC with the *KRAS G12C* mutation.[31] Given the limited sample size and lack of formal statistical inference, the efficacy of IBI351 in different subgroups should be interpreted with caution. The biomarker analysis in our study mainly focused on gene mutation; a more comprehensive analysis involving tumor mutation burden, PD-L1, and other novel biomarkers at transcription or protein levels could be informative in identifying patients who are more likely to benefit from IBI351. Molecular mechanisms underlying the acquired resistance of IBI351 are also yet to be clarified.

In conclusion, IBI351 monotherapy was found to have promising and durable efficacy and manageable safety profiles in previously treated NSCLC. Further clinical investigations of IBI351 in combination with or in comparison to other NSCLC treatments are warranted.

CRediT Authorship Contribution Statement

Qing Zhou: Conceptualization, Data curation, Writing - Original draft, Writing - review & editing.

Xiangjiao Meng: Conceptualization, Data curation, Writing - Original draft preparation, Writing - review & editing.

Longhua Sun: Data curation, Writing - review & editing.

Dingzhi Huang: Data curation, Writing - review & editing.

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Mengna Huang: Formal analysis, Writing - review & editing

Jiya Sun: Formal analysis, Writing - review & editing

Sujie Zhang: Data curation, Writing - review & editing

Hui Zhou: Conceptualization, Writing - review & editing

Yi-Long Wu: Supervision, Conceptualization, Data curation, Writing - Original draft preparation, Writing - review & editing.

Data Availability Statement

The de-identified data underlying this report can be made available upon reasonable request to the corresponding author.

Disclosure

Prof. Zhou reports honoraria from AstraZeneca, Lilly, Roche, Pfizer, Boehringer Ingelheim, MSD Oncology, Bristol-Myers Squibb, Hengrui. Prof. Wu reports honoraria from AstraZeneca, Lilly, Roche, Pfizer, Boehringer Ingelheim, MSD Oncology, Bristol-Myers Squibb, Hengrui; consulting or advisory role from AstraZeneca, Roche, Boehringer Ingelheim, Takeda; and research funding from Boehringer Ingelheim, Roche, Pfizer, Bristol-Myers Squibb. Dr. Huang, Dr. Huang, Dr. Sun, Dr. Zhang, and Dr. Zhou are employees of the sponsor Innovent Biologics (Suzhou) Co., Ltd. The remaining authors declare no conflict of interest.

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Supplementary Data

Note: To access the supplementary material accompanying this article, visit the online version of the *Journal of Thoracic Oncology* at www.jto.org and at https://10.1016/j.jtho.2024.08.005.

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